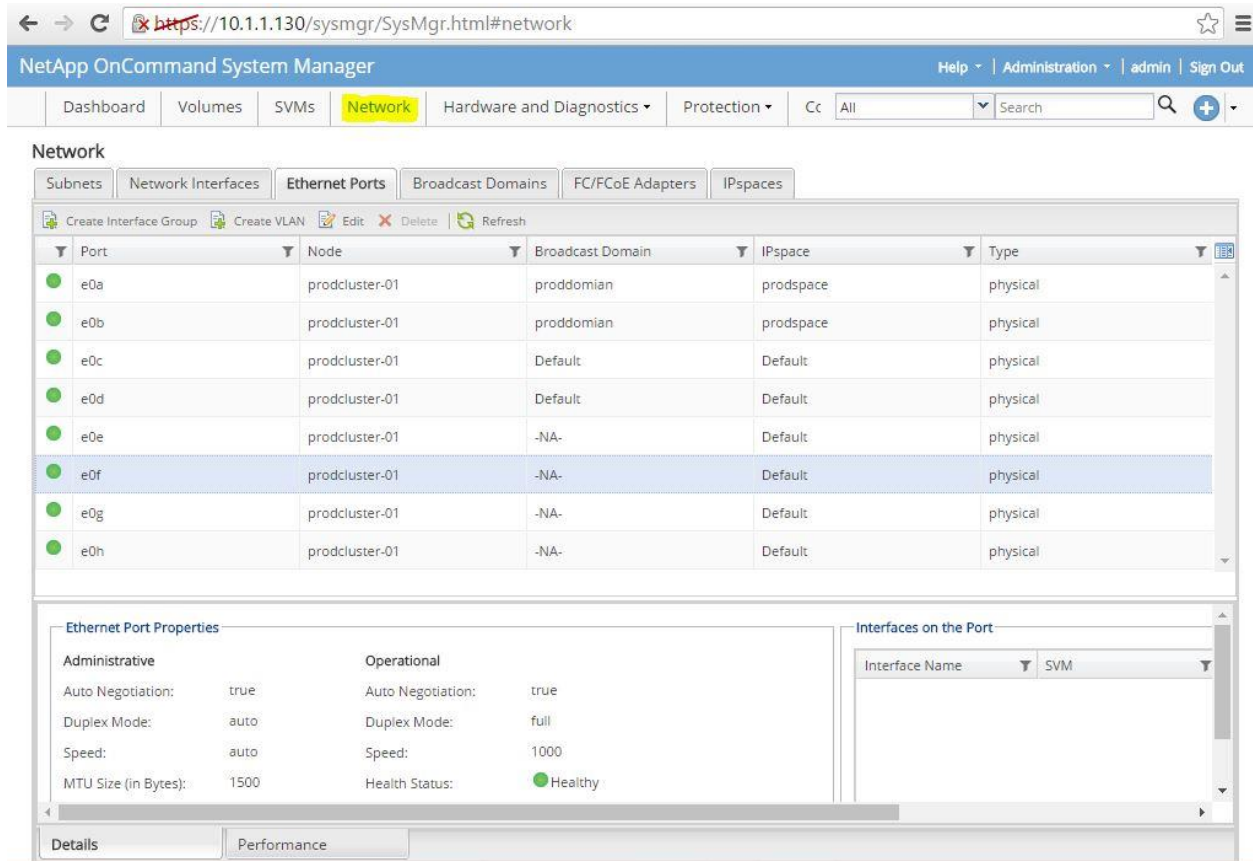


Network Management:

Listing the Network port using System Manager.



The screenshot shows the NetApp OnCommand System Manager web interface. The browser address bar displays `https://10.1.1.130/sysmgr/SysMgr.html#network`. The navigation bar includes links for Dashboard, Volumes, SVMs, Network (highlighted), Hardware and Diagnostics, Protection, and Cc. A search bar is also present. The main content area is titled 'Network' and contains several tabs: Subnets, Network Interfaces, Ethernet Ports (selected), Broadcast Domains, FC/FCoE Adapters, and IPspaces. Below the tabs, there is a table listing network interfaces. The table has columns for Port, Node, Broadcast Domain, IPspace, and Type. The 'e0f' interface is highlighted. Below the table, there are two sections: 'Ethernet Port Properties' and 'Interfaces on the Port'. The 'Ethernet Port Properties' section shows administrative and operational details for the selected interface. The 'Interfaces on the Port' section shows a list of interfaces connected to the selected port.

Port	Node	Broadcast Domain	IPspace	Type
e0a	prodcluster-01	proddomian	prodspace	physical
e0b	prodcluster-01	proddomian	prodspace	physical
e0c	prodcluster-01	Default	Default	physical
e0d	prodcluster-01	Default	Default	physical
e0e	prodcluster-01	-NA-	Default	physical
e0f	prodcluster-01	-NA-	Default	physical
e0g	prodcluster-01	-NA-	Default	physical
e0h	prodcluster-01	-NA-	Default	physical

Ethernet Port Properties

Administrative		Operational	
Auto Negotiation:	true	Auto Negotiation:	true
Duplex Mode:	auto	Duplex Mode:	full
Speed:	auto	Speed:	1000
MTU Size (in Bytes):	1500	Health Status:	Healthy

Interfaces on the Port

Interface Name	SVM
----------------	-----

Listing the network ports using CLI.

```
prodcluster::> net port show
(network port show)

Node: prodcluster-01
```

Port	IPspace	Broadcast Domain	Link	MTU	Speed (Mbps) Admin/Oper	Health Status
e0a	prodspace	proddomian	up	1500	auto/1000	healthy
e0b	prodspace	proddomian	up	1500	auto/1000	healthy
e0c	Default	Default	up	1500	auto/1000	healthy
e0d	Default	Default	up	1500	auto/1000	healthy
e0e	Default	-	up	1500	auto/1000	healthy
e0f	Default	-	up	1500	auto/1000	healthy
e0g	Default	-	up	1500	auto/1000	healthy
e0h	Default	-	up	1500	auto/1000	healthy

```
8 entries were displayed.

prodcluster::>
```

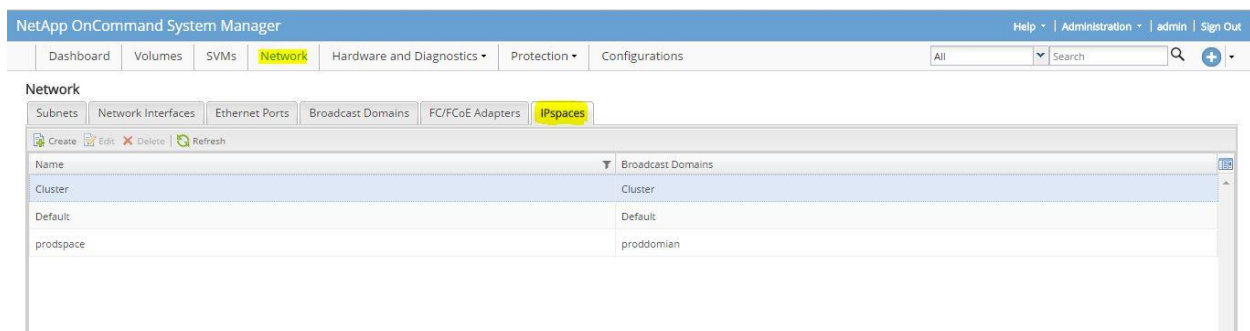
Listing the IPspace information.

```
prodcluster::>
prodcluster::> net ipspace show
(network ipspace show)
```

IPspace	Vserver List	Broadcast Domains
Cluster		
Default	Cluster	Cluster
prodspace	prodcluster	Default
	prodspace, vs1	proddomian

```
3 entries were displayed.

prodcluster::>
```



Creating a new IPspace.

```

prodcluster::>
prodcluster::> net ipspace create dept2
(network ipspace create)

prodcluster::>
prodcluster::> net ipspace show
(network ipspace show)

```

IPspace	Vserver List	Broadcast Domains
Cluster	Cluster	Cluster
Default	prodcluster	Default
dept2	dept2	-
prodspace	prodspace, vs1	proddomian

```

4 entries were displayed.

prodcluster::>
prodcluster::>

```

Creating an ifgrp for network port redundancy.

```

prodcluster::>
prodcluster::> net port ifgrp create -node prodcluster-01 -ifgrp a0a -distr-func
ip -mode singlemode
(network port ifgrp create)

prodcluster::>
prodcluster::> net port ifgrp create -node prodcluster-01 -ifgrp ala -distr-func
ip -mode singlemode
(network port ifgrp create)

prodcluster::>
prodcluster::> net port ifgrp show
(network port ifgrp show)

```

Node	Port IfGrp	Distribution Function	MAC Address	Active Ports	Ports
prodcluster-01	a0a	ip	02:0c:29:63:54:22	none	-
	ala	ip	02:0c:29:63:54:2c	none	-

```

2 entries were displayed.

prodcluster::>

```

Add ports to an ifgrp.

```

10.1.1.130 - PuTTY
prodcluster::> net port ifgrp add-port -node prodcluster-01 -ifgrp a0a -port e0e ^
(network port ifgrp add-port)

prodcluster::> net port ifgrp add-port -node prodcluster-01 -ifgrp a0a -port e0f
(network port ifgrp add-port)

prodcluster::> net port ifgrp add-port -node prodcluster-01 -ifgrp ala -port e0g
(network port ifgrp add-port)

prodcluster::> net port ifgrp add-port -node prodcluster-01 -ifgrp ala -port e0h
(network port ifgrp add-port)

prodcluster::>
prodcluster::> net port ifgrp show
(network port ifgrp show)

```

Node	Port	Distribution	MAC Address	Active	Ports
	IfGrp	Function		Ports	
prodcluster-01	a0a	ip	02:0c:29:63:54:22	partial	e0e, e0f
	ala	ip	02:0c:29:63:54:2c	partial	e0g, e0h

```

2 entries were displayed.
prodcluster::>

```

```

prodcluster::> net port show
(network port show)

Node: prodcluster-01

```

Port	IPspace	Broadcast Domain	Link	MTU	Speed (Mbps)	Health
					Admin/Oper	Status
a0a	Default	-	up	1500	auto/1000	healthy
ala	Default	-	up	1500	auto/1000	healthy
e0a	prodspace	proddomian	up	1500	auto/1000	healthy
e0b	prodspace	proddomian	up	1500	auto/1000	healthy
e0c	Default	Default	up	1500	auto/1000	healthy
e0d	Default	Default	up	1500	auto/1000	healthy
e0e	Default	-	up	1500	auto/1000	healthy
e0f	Default	-	up	1500	auto/1000	healthy
e0g	Default	-	up	1500	auto/1000	healthy
e0h	Default	-	up	1500	auto/1000	healthy

```

10 entries were displayed.
prodcluster::>

```

Listing the Broadcast Domain information.

```

prodcluster::> net port broadcast-domain show
(network port broadcast-domain show)
IPspace Broadcast
Name      Domain Name      MTU  Port List
-----
Cluster Cluster          9000
Default Default          1500
prodspace
  proddomian          1500
3 entries were displayed.
prodcluster::>
prodcluster::>

```

Name	Domain Name	MTU	Port List	Update Status	Details
Cluster	Cluster	9000	-	-	
Default	Default	1500	prodcluster-01:e0c	complete	
			prodcluster-01:e0d	complete	
prodspace	proddomian	1500	prodcluster-01:e0a	complete	
			prodcluster-01:e0b	complete	

Creating a new Broadcast Domain.

```
10.1.1.130 - PuTTY

prodcluster::> net port broadcast-domain create -ip-space dept2 -broadcast-domain
dept2domain -mtu 1500 -ports prodcluster-01:a0a
(network port broadcast-domain create)

prodcluster::> net port broadcast-domain show
(network port broadcast-domain show)
IPspace Broadcast
Name      Domain Name      MTU      Port List      Update
-----
Cluster Cluster      9000      -              -
Default Default      1500      prodcluster-01:e0c complete
prodcluster-01:e0d complete
dept2     dept2domain      1500      prodcluster-01:a0a complete
prodspace
         proddomain  1500      prodcluster-01:e0a complete
         prodcluster-01:e0b complete
4 entries were displayed.

prodcluster::>
```

Adding ports to an existing Broadcast Domain.

```
prodcluster::> net port broadcast-domain add-ports -broadcast-domain dept2domain
-ports prodcluster-01:a1a -ip-space dept2
(network port broadcast-domain add-ports)

prodcluster::> net port broadcast-domain show
(network port broadcast-domain show)
IPspace Broadcast
Name      Domain Name      MTU      Port List      Update
-----
Cluster Cluster      9000      -              -
Default Default      1500      prodcluster-01:e0c complete
prodcluster-01:e0d complete
dept2     dept2domain      1500      prodcluster-01:a0a complete
prodcluster-01:a1a complete
prodspace
         proddomain  1500      prodcluster-01:e0a complete
         prodcluster-01:e0b complete
4 entries were displayed.
```

```

prodcluster::>
prodcluster::> net port show
(network port show)

Node: prodcluster-01

```

Port	IPspace	Broadcast Domain	Link	MTU	Speed (Mbps) Admin/Oper	Health Status
a0a	dept2	dept2domain	up	1500	auto/1000	healthy
ala	dept2	dept2domain	up	1500	auto/1000	healthy
e0a	prodspace	proddomain	up	1500	auto/1000	healthy
e0b	prodspace	proddomain	up	1500	auto/1000	healthy
e0c	Default	Default	up	1500	auto/1000	healthy
e0d	Default	Default	up	1500	auto/1000	healthy
e0e	Default	-	up	1500	auto/1000	healthy
e0f	Default	-	up	1500	auto/1000	healthy
e0g	Default	-	up	1500	auto/1000	healthy
e0h	Default	-	up	1500	auto/1000	healthy

```

10 entries were displayed.

prodcluster::>
prodcluster::>

```

Listing and creating subnets.

```

prodcluster::>
prodcluster::> net subnet show
(network subnet show)

IPspace: prodspace

```

Subnet Name	Subnet	Broadcast Domain	Gateway	Avail/ Total	Ranges
prodspace	10.0.0.0/8	proddomain	10.0.0.1	8/10	10.2.2.121-10.2.2.130

```

prodcluster::>
prodcluster::> net subnet create -ip-space dept2 -subnet-name dept2sub -broadcast
-domain dept2domain -subnet 10.0.0.0/8 -gateway 10.0.0.1 -ip-ranges 10.5.5.121-1
0.5.5.140
(network subnet create)

prodcluster::>
prodcluster::> net subnet show
(network subnet show)

IPspace: dept2

```

Subnet Name	Subnet	Broadcast Domain	Gateway	Avail/ Total	Ranges
dept2sub	10.0.0.0/8	dept2domain	10.0.0.1	20/20	10.5.5.121-10.5.5.140

Ifgrp port status.

```
10.1.1.130 - PuTTY
prodcluster::> net port ifgrp show -node prodcluster-01 -ifgrp a0a -fields ports
(network port ifgrp show)
node          ifgrp ports
-----
prodcluster-01 a0a    e0e,e0f

prodcluster::>
prodcluster::>
prodcluster::> net port ifgrp show -node prodcluster-01 -ifgrp a0a -fields up-po
rts
(network port ifgrp show)
node          ifgrp up-ports
-----
prodcluster-01 a0a    e0e

prodcluster::> net port ifgrp show -node prodcluster-01 -ifgrp a0a -fields down-
ports
(network port ifgrp show)
node          ifgrp down-ports
-----
prodcluster-01 a0a    e0f

prodcluster::>
```

Creating a SVM LIF (Logical Interface).

```
10.1.1.130 - PuTTY
prodcluster::>
prodcluster::>
prodcluster::> net int create -vserver dept2vs2 -lif dept2-NAS -role data -data-
protocol cifs,nfs,fcache -home-node prodcluster-01 -home-port a0a -subnet-name d
ept2sub -status-admin up
(network interface create)

prodcluster::>
prodcluster::>
prodcluster::>
prodcluster::> net int show
(network interface show)
Vserver      Logical   Status   Network      Current      Current  Is
Interface    Admin/Oper  Address/Mask Node         Port        Home
-----
dept2vs2     dept2-NAS   up/up    10.5.5.121/8 prodcluster-01 a0a        true
prodcluster  cluster_mgmt up/up    10.1.1.130/8 prodcluster-01 e0c        true
prodcluster-01_mgmt1 up/up    10.1.1.131/8 prodcluster-01 e0d        true
```

Pinging the Network ip address.

```
prodcluster::>
prodcluster::> net ping -lif dept2-NAS -vserver dept2vs2 -destination 10.5.5.150
(network ping)
10.5.5.150 is alive

prodcluster::> net ping -lif dept2-NAS -vserver dept2vs2 -destination 10.5.5.121
(network ping)
10.5.5.121 is alive

prodcluster::>
prodcluster::>
prodcluster::>
```

```
10.1.1.130 - PuTTY
prodcluster::> net int show
(network interface show)
```

Vserver	Logical Interface	Status Admin/Oper	Network Address/Mask	Current Node	Current Port	Is Home
dept2vs2	dept2-NAS	up/up	10.5.5.121/8	prodcluster-01	a0a	true
	dept2-SAN	up/up	10.5.5.150/8	prodcluster-01	a0a	true
prodcluster	cluster_mgmt	up/up	10.1.1.130/8	prodcluster-01	e0c	true
	prodcluster-01_mgmt1	up/up	10.1.1.131/8	prodcluster-01	e0d	true
vs1	data-nas	up/up	10.2.2.122/8	prodcluster-01	e0a	true
	vs1_admin_lif1	up/up	10.2.2.121/8	prodcluster-01	e0a	true

6 entries were displayed.

Disable the port to test.

```
10.1.1.130 - PuTTY
prodcluster::>
prodcluster::>
prodcluster::> net port show
(network port show)

Node: prodcluster-01
```

Port	IPspace	Broadcast Domain	Link	MTU	Speed (Mbps) Admin/Oper	Health Status
a0a	dept2	dept2domain	up	1500	auto/1000	healthy
ala	dept2	dept2domain	up	1500	auto/1000	healthy
e0a	prodspace	proddomian	up	1500	auto/1000	healthy
e0b	prodspace	proddomian	up	1500	auto/1000	healthy
e0c	Default	Default	up	1500	auto/1000	healthy
e0d	Default	Default	up	1500	auto/1000	healthy
e0e	Default	-	up	1500	auto/1000	healthy
e0f	Default	-	up	1500	auto/1000	healthy
e0g	Default	-	up	1500	auto/1000	healthy
e0h	Default	-	up	1500	auto/1000	healthy

```
10 entries were displayed.

prodcluster::>
prodcluster::>
prodcluster::>
```

Port status is down now.

```
prodcluster::>
prodcluster::>
prodcluster::> net port show
(network port show)

Node: prodcluster-01
```

Port	IPspace	Broadcast Domain	Link	MTU	Speed (Mbps) Admin/Oper	Health Status
a0a	dept2	dept2domain	up	1500	auto/1000	healthy
ala	dept2	dept2domain	up	1500	auto/1000	healthy
e0a	prodspace	proddomian	up	1500	auto/1000	healthy
e0b	prodspace	proddomian	up	1500	auto/1000	healthy
e0c	Default	Default	up	1500	auto/1000	healthy
e0d	Default	Default	up	1500	auto/1000	healthy
e0e	Default	-	down	1500	auto/1000	-
e0f	Default	-	up	1500	auto/1000	healthy
e0g	Default	-	up	1500	auto/1000	healthy
e0h	Default	-	up	1500	auto/1000	healthy

```
10 entries were displayed.

prodcluster::>
```

Still it is pinging using another survival port.

```
10.1.1.130 - PuTTY
prodcluster::> net ping -lif dept2-NAS -vserver dept2vs2 -destination 10.5.5.121
(network ping)
10.5.5.121 is alive

prodcluster::> net ping -lif dept2-NAS -vserver dept2vs2 -destination 10.5.5.150
(network ping)
10.5.5.150 is alive

prodcluster::>
prodcluster::> net port ifgrp show -node prodcluster-01 -ifgrp a0a -fields up-po
rts
(network port ifgrp show)
node          ifgrp up-ports
-----
prodcluster-01 a0a    e0f

prodcluster::> net port ifgrp show -node prodcluster-01 -ifgrp a0a -fields down-
ports
(network port ifgrp show)
node          ifgrp down-ports
-----
prodcluster-01 a0a    e0e
```

Creating a VLAN.

```
prodcluster::> net port vlan create -node prodcluster-01 -vlan-name ala-10
(network port vlan create)

prodcluster::>
prodcluster::> network port vlan show
Network Network
Node  VLAN Name Port  VLAN ID  MAC Address
-----
prodcluster-01
      ala-10  ala    10      02:0c:29:63:54:2c

prodcluster::>
prodcluster::>
prodcluster::>
```